

ASSIGNMENT 1

PROBLEM STATEMENT: -

Set up and Configuration of Hadoop Using CloudEra/ Google Cloud BigQuery. Databricks Lakehouse Platform. Snowflake, Amazon Redshift

OBJECTIVE:

1. Set up a multi-node Hadoop cluster using Cloudera Manager And Configure Hadoop Distributed File System (HDFS) for distributed storage.
2. Create a Google Cloud project and enable BigQuery API and Configure billing and access controls.
3. Create a Databricks workspace and configure clusters.
4. Create a Snowflake account and configure access controls.
5. Launch and configure an Amazon Redshift cluster.

Software Requirements:

Cloudera Manager Server, Google Cloud Platform account and SDK for command-line tools , Databricks account and Apache Spark environment , Snowflake account and UI access , Amazon Redshift cluster , AWS account with permissions and JDBC/ODBC driver.

Hardware Requirements:

- Network connectivity, Storage for Hadoop Distributed File System (HDFS).
- Amazon Redshift, Snowflake is a cloud-based service, Databricks are provides a managed environment, no specific hardware requirements.

THEORY:

Setting up and configuring Hadoop using Cloudera, Google Cloud BigQuery, Databricks Lakehouse Platform, Snowflake, and Amazon Redshift involves different procedures for each platform. Below, I'll provide a brief overview and the general steps for setting up each one. Please note that detailed setup and configuration steps can vary based on the specific version and updates of these platforms.

• Hadoop using Cloudera:

1. **Cloudera Distribution:** Download and install Cloudera Distribution for Hadoop (CDH) from the Cloudera website.

2. **Cloudera Manager:** Use Cloudera Manager to set up and manage your Hadoop cluster. Follow the installation and configuration wizard in Cloudera Manager to define hosts, services, and configuration parameters.
3. **HDFS and MapReduce:** Configure Hadoop Distributed File System (HDFS) and MapReduce based on your requirements.
4. **Cluster Deployment:** Deploy the configured cluster.

- **Google Cloud Big Query:**

1. **Google Cloud Console:** Log in to the [Google Cloud Console](<https://console.cloud.google.com/>).
2. **Enable Big Query API:** In the GCP Console, navigate to "APIs & Services" > "Library." Enable the Big Query API.
3. **Create a Dataset:** Create a BigQuery dataset to organize your tables.
4. **Create Tables:** Create tables within your dataset, defining the schema and specifying data sources.
5. **Access Control:** Configure access control for datasets and tables.

- **Databricks Lakehouse Platform:**

1. **Databricks Account:**

Sign up for a Databricks account: [Databricks](<https://databricks.com/trydatabricks>).

2. **Create a Workspace:** Create a Databricks workspace. Configure clusters, libraries, and storage settings.
3. **Connect to Data Sources:** Connect Databricks to your data sources, such as storage accounts or databases

- **Snowflake:**

1. **Snowflake Account:**

Sign up for a Snowflake account: [Snowflake](<https://www.snowflake.com/free-trial/>).

2. **Snowflake UI:** Use the Snowflake web UI to configure your account, create warehouses, databases, and schemas.
3. **Connectivity:** Configure network policies and connectivity settings.
4. **User and Role Setup:** Set up users and roles with appropriate privileges.

• Amazon Redshift:

1. **AWS Account:** Sign up for an AWS account: [AWS Signup](<https://aws.amazon.com/>).
2. **Amazon Redshift Cluster:** In the AWS Management Console, navigate to Amazon Redshift. Create a new Redshift cluster.
3. **Configure Redshift:** Set up databases, tables, and users as needed. Configure security groups and network settings.

These are general steps, and each platform has detailed documentation that you should follow for the specific versions you are using. Ensure that you follow best practices for security, scalability, and performance in each case.

Applications –**• Hadoop Using Cloudera :**

1. Data Warehousing.
2. Batch Processing.
3. Machine Learning.

• Google Cloud BigQuery :

1. Ad Hoc Analytics
2. Real-time Analytics

• Databricks Lakehouse Platform:

1. Unified Analytics.
2. ETL and Data Integration.
3. Data Exploration and Visualization.

• Snowflake:

1. Cloud Data Warehousing
2. Data Sharing
3. Time Travel Queries.

• Amazon Redshift:

1. Business Intelligence (BI).
2. Data Integration.

Setup - Setting up and configuring Hadoop, Google Cloud BigQuery, Databricks Lakehouse Platform, Snowflake, and Amazon Redshift involves different procedures for each platform. Below, I'll provide you with a brief overview and the steps for setting up each one. Keep in mind that detailed setup and configuration steps can vary based on the specific version and updates of these platforms.

- **Hadoop Setup:**

- Download and Install Hadoop:**

- Visit the Apache Hadoop website and download the version you want to use: [Hadoop

- Downloads](<http://hadoop.apache.org/releases.html>) Follow the installation instructions provided with the Hadoop distribution.

- Configure Hadoop:**

- Modify the 'core-site.xml', 'hdfs-site.xml', and other configuration files as needed. Configure the Hadoop environment variables.

- Start Hadoop Services:**

- Use the 'start-dfs.sh' and 'start-yarn.sh' scripts to start the Hadoop Distributed File System (HDFS) and Yet Another Resource Negotiator (YARN).

- **Google Cloud BigQuery Setup:**

- 1. Create a Google Cloud Platform (GCP) Project:**

- Go to the [Google Cloud Console](<https://console.cloud.google.com/>). Create a new project.

- 2. Enable BigQuery API:**

- In the GCP Console, navigate to "APIs & Services" > "Library". Enable the BigQuery API.

- 3. Set Up Authentication:**

- Create service account credentials with BigQuery access. Download the JSON key file and set the 'GOOGLE_APPLICATION_CREDENTIALS' environment variable.

- **Databricks Lakehouse Platform Setup:**

- 1. Create a Databricks Account:**

- Sign up for a Databricks account: [Databricks](<https://databricks.com/try-databricks>)

- 2. Configure Databricks Workspace:**

- Create a Databricks workspace. Configure clusters, libraries, and other settings as needed.

3. Connect to Data Sources: Connect Databricks to your data sources, such as storage accounts or databases.

• **Snowflake Setup:**

1. Create a Snowflake Account: Sign up for a Snowflake account: [Snowflake](<https://www.snowflake.com/free-trial/>)

2. Configure Snowflake: Create a Snowflake warehouse, database, and schema Set up users and roles.

3. Connect to Snowflake: Use a Snowflake client or integration tool to connect to Snowflake.

• **Amazon Redshift Setup:**

1. Create an AWS Account: Sign up for an AWS account: [AWS Signup] (<https://aws.amazon.com/>)

2. Create an Amazon Redshift Cluster: In the AWS Management Console, navigate to Amazon Redshift. Create a new Redshift cluster.

3. Configure Redshift: Set up databases, tables, and users as needed. Configure security groups and network settings.

Each platform has its own documentation with detailed setup and configuration instructions. Follow the official documentation for the specific versions you are using. Additionally, consider security best practices and compliance requirements when setting up these platforms.

CONCLUSION:

In this way we have Set up and Configuration Hadoop Using CloudEra/ Google Cloud BigQuery, Databricks Lakehouse Platform, Snowflake, Amazon Redshift.

ORAL QUESTION:

1. How did you configure and set up Hadoop within the CloudEra/Google Cloud BigQuery environment?
2. What are the key advantages of using CloudEra or Google Cloud BigQuery for hosting Hadoop?
3. How did you ensure data integrity and security within this environment?
4. What are the key features of Databricks Lakehouse Platform that enhance data processing and analytics capabilities?